

MCQ MASTERCLASS: SMART MATHS

Quantitative Aptitude & Arithmetic Shortcuts – 2026 Edition



What this Chapter is REALLY About

Think of this chapter as “**numbers behaving smartly in real life**”:

- **Ratios** → comparing things
- **Percentages** → parts out of 100
- **Interest** → money growing 🌱
- **Profit/Loss** → shopkeeper math
- **Data numbers** → quick calculations

👉 **MCQs are NOT about long solving. They test recognition + shortcut thinking.**



GOLDEN RULE (Most Important)

✅ **Never solve fully first.**

➡ **Estimate → Eliminate → Then calculate.**

70% of MCQs can be solved without full calculation.



1 Percentage → Convert to FRACTION instantly

Percentage	Fraction Trick
50%	$\frac{1}{2}$
25%	$\frac{1}{4}$
20%	$\frac{1}{5}$
12.5%	$\frac{1}{8}$
33.33%	$\frac{1}{3}$
66.66%	$\frac{2}{3}$

Example: 20% of 450 = $\frac{1}{5} \times 450 = 90$ (no calculator needed)

EXAM HACK If options are far apart → mental fraction method wins.

2 Profit & Loss MAGIC FORMULA

Instead of remembering many formulas, remember **ONE**:

$$\text{Profit\%} = (\text{Profit} / \text{CP}) \times 100$$

Fast Trick: If Profit = 20%, Selling Price = 120% of CP. So, **SP = 1.2 × CP**. No long solving needed.

3 Successive Percentage Trick 🔥

If increase = a% then decrease = b%, use:

$$\text{Net\%} = a + b + (ab / 100)$$

⚠️ **Add sign carefully (+/-).**

Example: +20% then -10% → $20 - 10 - (20 \times 10) / 100 = 10 - 2 = 8\% \text{ increase}$. (Done in 5 seconds)

4 Simple Interest vs Compound Interest Shortcut

Key Observation Trick: If time = 2 years,

$$\text{CI} - \text{SI} = (P \times R^2) / 100^2$$

No full compound formula needed 😎

5 Ratio Questions — SMART SCALING

Example: Ratio = 3 : 5, Total = 80.

Add ratio parts: $3 + 5 = 8$.

One part = $80 \div 8 = 10$.

Numbers = 30 and 50 ✓

6 Option Substitution Trick (Topper Secret ★)

If options are numbers, put option into question instead of solving.

Example: "Number increased by 20% becomes 72"

Options: 50, 60, 65, 70

Check quickly: $60 + 20\% = 60 + 12 = 72$ ✓ (Solved in 3 seconds)

7 Unit Digit Trick (Huge Time Saver)

If options differ at last digit: $23 \times 47 \rightarrow 3 \times 7 = 21 \rightarrow$ last digit **1**. Only option ending with 1 is correct.

8 Approximation Method (Exam King 🏰)

Round numbers slightly: $49 \times 19 \approx 50 \times 20 = 1000$. Choose closest option.

9 Interest Growth Shortcut

If rate = 10%, each year multiply by 1.1.
After 2 years: $P \times 1.1 \times 1.1$. Super fast mental math.

10 Elimination Rule (Most Powerful)

✗ Remove impossible options: **negative value? too big? too small?** Often reduces 4 options to 2 instantly. Probability of correct answer doubles!

5-SECOND MCQ STRATEGY

- 1 Read question
- 2 Guess approximate answer
- 3 Eliminate wrong options

- 4 Apply shortcut
- 5 Calculate only if needed

TOPPER EXAM MANTRA

👉 MCQs reward speed thinking, not long maths.

If solution taking > 20 sec → you're doing it wrong.

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